



Big Data & Data Mining Project Report

Road Traffic Accident Analysis & Social Network Analysis

Module: 772238 – Big Data & Data Mining
MSc Artificial Intelligence & Data Science

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This report is submitted in accordance with academic integrity policies.

Abstract

This report presents an analytical investigation into **UK road traffic accidents in 2019** and a structural assessment of a **Facebook social network** derived from SNAP data. Accident records were mined using various **data analysis and data mining techniques**, including **temporal analysis**, **association rule mining** with the **Apriori** algorithm, **clustering**, **spatiotemporal mapping**, and **time series forecasting**. The results indicate clear **temporal peaks** in accident occurrence, heterogeneous behaviour across **different motorcycle categories**, significant **risk periods for pedestrians**, and geographically concentrated **hotspot zones** within Hull and the East Riding. Predictive models demonstrated effective **weekly accident forecasting** for selected police regions and high-incident **Lower Super Output Areas**. Apriori rules revealed influential **environmental and road-condition factors** associated with accident severity. The **social network analysis** explores network structure, **centrality distribution**, and **community detection** through multiple algorithms. Overall, the results support **policy recommendations** that inform actionable enhancements in road safety and provide broader insights into social network behaviour.

Part (a) – Road Traffic Accident Data Analysis & Policy Recommendations

0.1 Introduction

This report evaluates **road traffic accidents across the United Kingdom during 2019**, using the `accident_data_v1.0.0_2023.db` database. The dataset consists of approximately **117,536 recorded accidents**, reported across **51 police forces** and covering **33,748 unique geographic codes**, providing extensive national coverage and a robust basis for high-resolution analysis. The aim of this study is to identify patterns in the occurrence and outcomes of accidents, determine key **temporal, spatial, and contextual factors** associated with high-risk incidents, and propose **evidence-based recommendations** to support policy and safety interventions for the UK **Department for Transport**. A range of **data mining techniques** are applied, including **time series analysis, Apriori association rule mining, clustering, spatio-temporal mapping, and forecasting models**, to analyse when, where, and under what conditions accidents are most likely to occur. The level of analysis extends to **predictive modelling**, interpretation, and policy recommendations designed to reduce the **frequency and severity** of accidents.

0.2 Analysis & Results

0.2.1 When Accidents Occur: Time-of-Day & Day-of-Week Patterns

The analysis on frequency of accident in 2019 depicts a distinct patterns associated with both the day of the week and time period of the day. Accident occurrence particularly peaks during conventional commuting periods. The highest volume recorded is **10,198**, occurred at **17:00**. Further, peaks at **16:00–18:00** and **08:00–09:00**, respectively at evening and morning rush hours. In contrary, the least frequency of accident occurred at **04:00**, where around **791 incidents** were recorded. This shows the lower volume of travel in the dawn. Accidents during the peak travelling hours of **08:00–09:00 and 16:00–18:00** average **8,326.2**, which is twice more than that of the normal hours average of **3,995.0**.

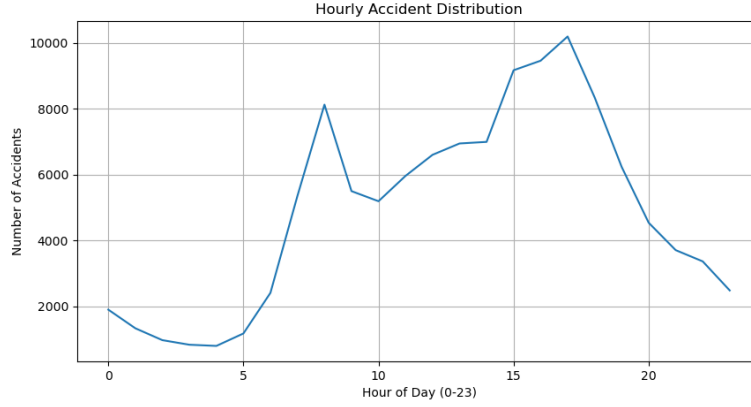


Figure 1: Hourly Accident Distribution in 2019

A notable difference between accident on weekdays and weekend is observed in weekly patterns. **19,056** is the maximum number of accidents occurred on Day **5 (Friday)**. This indicates increased traffic levels during end-of-week. On the other hand, least volume of accidents occur on **Sunday (13,245 incidents)**. Making the difference between the weekday and weekend averages (**17,809.0** vs. **14,245.5**, respectively) is of **3,563.5**. These trends may indicate factor like commercial and industrial commuting activity during weekdays. An increased recreational travel and possible drunk and drive during weekend evenings.

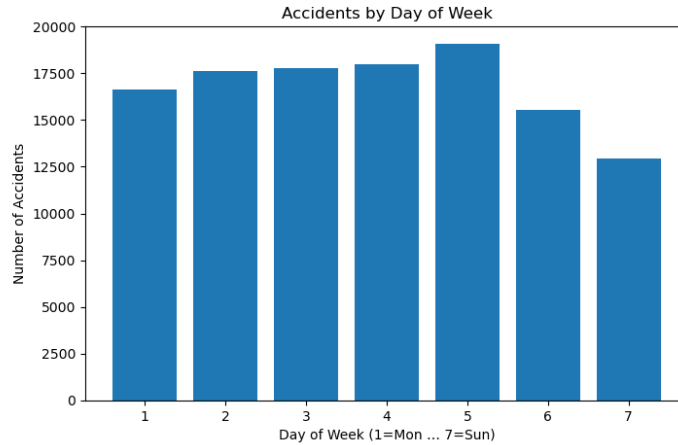


Figure 2: Accidents by Day of Week in 2019

This is visually reinforced by the combination of day and hour interactions in the heatmap, whereby dense concentrations of accidents are seen around

08:00–09:00 and **16:00–18:00** throughout the weekdays, hence showing strong temporal dependence. These time-based risk patterns provide a vital basis for targeted intervention strategies, including police deployments, traffic light optimization, public awareness campaigns, and variable speed limit enforcement.

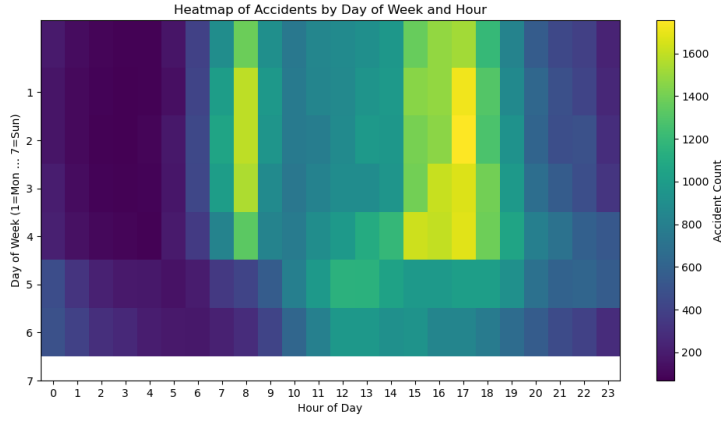


Figure 3: Heatmap of Accidents by Day of Week and Hour

0.2.2 Motorcycle Accident Temporal Patterns (125cc vs 125–500cc vs >500cc)

The time-focused assessment of motorcycle accidents divided by capacity of engine ($\leq 125\text{cc}$, $125\text{--}500\text{cc}$, and $>500\text{cc}$) unfolds notable action-related and variations in usage among rider groups. In a total of **161,423 motorcycle accidents** almost **149,918 incidents** were recorded with the $>500\text{cc}$ category in 2019. Which is the maximum comparing to **8,495** in the $\leq 125\text{cc}$ group and **822** in the $125\text{--}500\text{cc}$ group. Across all categories, accident peaks consistently occur at **17:00**. This highlights higher concentration of travels in the evening, consistent with research identifying increased collision risk during darker travel periods [8]. Minimal activity is detected from the metrics observed at **03:00** for $\leq 125\text{cc}$ and $125\text{--}500\text{cc}$, and **04:00** for $>500\text{cc}$.

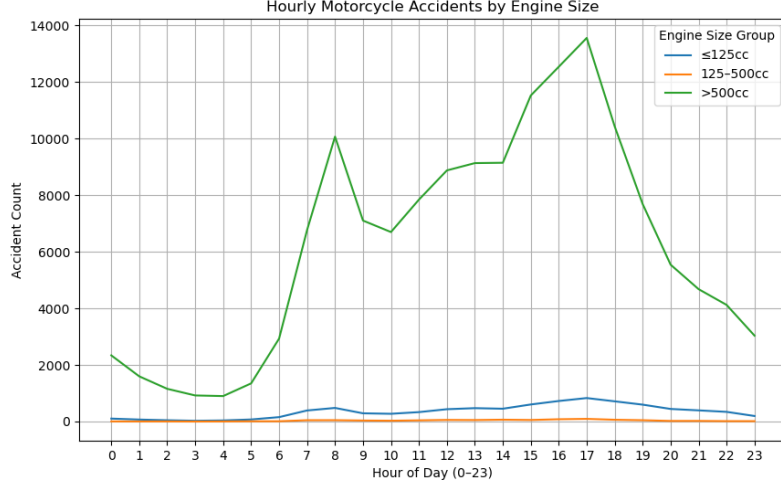


Figure 4: Hourly Motorcycle Accident Distribution by Engine Size

Daily patterns depict difference in peak accident days. **$\leq 125\text{cc}$ and $>500\text{cc}$ peak on Day 2 (Tuesday)**, whereas **$125\text{--}500\text{cc}$ peaks on Day 6 (Saturday)**. Weekday–weekend comparisons further highlight usage–based divergence: the average weekday accidents for **$>500\text{cc}$** are **8,414.4**, slightly higher than its weekend average of **8,372.0**. For **$\leq 125\text{cc}$** , weekday and weekend averages remain closely aligned (**476.6 vs 482.5**). The difference between the largest and smallest engine segments is **149,096 accidents**, indicating significant disparity in travel volume, which aligns with studies identifying variability in motorcycling–related risk patterns across different temporal contexts [2].

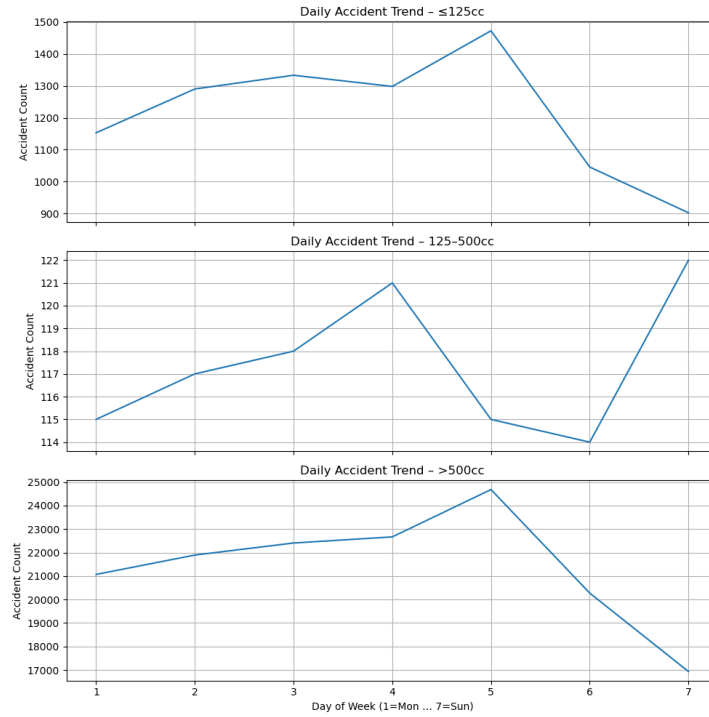


Figure 5: Daily Motorcycle Accident Pattern Comparison by Engine Capacity

Rush-hour behaviour is also strongly differentiated. $>500\text{cc}$ motorcycles record a rush-hour average of **10,737.8**, substantially higher than the non-rush-hour average of **5,064.7**, suggesting increased recreational and leisure traffic intensity. In contrast, $\leq 125\text{cc}$ motorcycles show stronger commuting utility with an average of **608.4** during peak travel hours compared to **287.0** off-peak. These patterns provide insight into demographic travel behaviours that influence exposure to accident risk.

0.2.3 Pedestrian Accident Timing Analysis

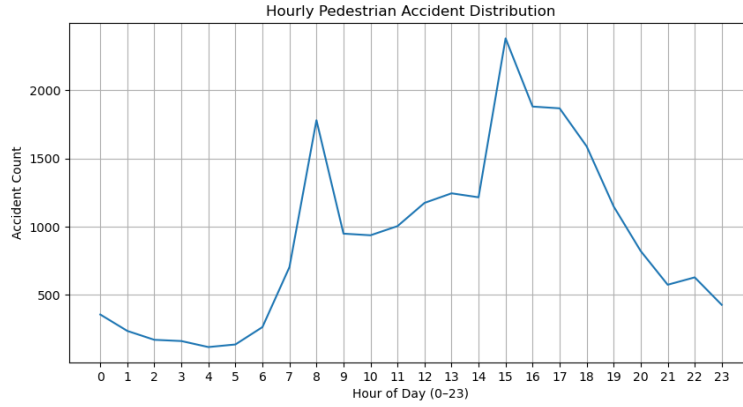


Figure 6: Hourly Pedestrian Accident Distribution

The chronological analysis of accidents involving pedestrian depicts time-related trends linked to commuting activity. Totally **21,770 accidents** involving pedestrians were recorded, the minimum of **119** such accidents occurs at **04:00 hour**. IN 2019, this peaks at **15:00 with 2,378 accidents**. Between **16:00–18:00** due to work and school related commute there is high volume of accidents observed. **11.27%** of accidents are recorded during night (**20:00–03:00**). Major reasons for this might be reduced visibility and sleep deprivation [8].

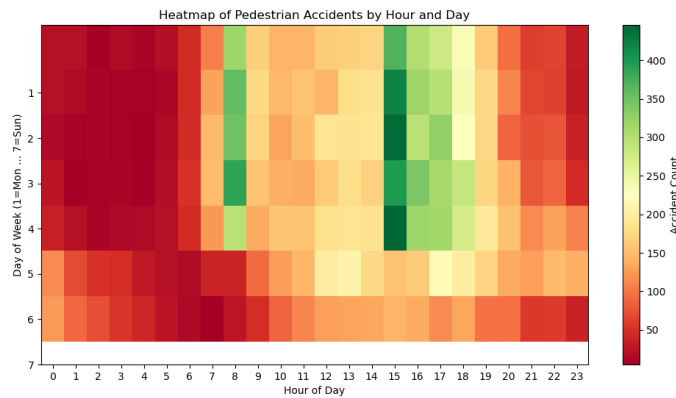


Figure 7: Pedestrian Accident Heatmap: Hour vs Day

Daily frequency depicts **1,415 incidents**, that is the maximum accidents

occurred on **Day 2 (Tuesday)**. The similarity in weekday and weekend averages (**1,253.6 vs 1,221.0**) shows consistent pedestrian-at-risk frequency. **1,612.2** is the average accidents in peak-hours. This is more than double that in normal-hour average (**721.5**). Severity analysis depicts **2.1% fatal, 26.3% serious and 71.5% slight accidents**. The heatmap affirms dense clustering during **08:00–09:00 and 15:00–18:00**, consistent with European pedestrian collision trends. Finding crucial periods for aimed traffic and pedestrian safety measures.

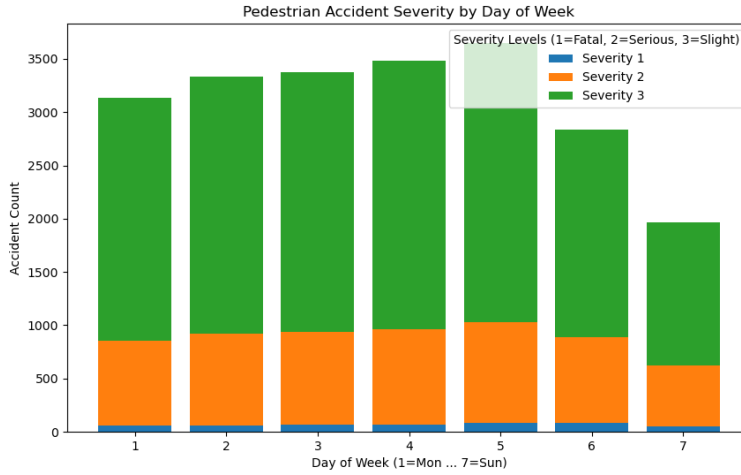


Figure 8: Pedestrian Accident Severity by Day

0.2.4 Association Rule Mining: Variables Influencing Severity (Apriori)

Association Rule Mining using the **Apriori** algorithm was applied to identify combinations of environmental and contextual conditions associated with accident severity outcomes. Categorical variables derived from the accident dataset were used to enable Apriori rule extraction without excessive computational overhead. Minimum thresholds of **support = 0.01**, **confidence = 0.30**, and **lift = 1.20** were selected to retain rules that were both statistically and practically meaningful. The algorithm initially generated **106,951 rules**, which reduced to **54,868** following threshold filtering, demonstrating substantial pruning of weak associations.

A summary of key rules related to severity is presented in **Table 1**. Notable patterns emerging from the rule set indicate that lighting conditions, time of day and environmental context strongly influence severity likelihood.

Rules involving evening and night travel under low visibility conditions consistently demonstrate high lift values, highlighting increased severity risk during periods of reduced visibility, a finding consistent with evidence regarding lighting and accident risk [8]. Vehicle-type influences and wet surface interactions were also associated with elevated serious injury outcomes, aligning with research demonstrating the importance of diverse factor interaction in crash severity analysis [4]. These findings provide evidence that severity outcomes are highly dependent on combined situational risk factors rather than single-variable effects, echoing conclusions in wider road-safety studies [2].

While Apriori provides interpretable rule relationships, limitations include reliance on categorical binning, sensitivity to data imbalance and lack of behavioural or demographic variables. Future refinement could incorporate **FP-Growth**, **decision trees**, or **logistic regression** to enhance predictive capability while integrating continuous features. Advanced sensor-based modelling has also shown potential in improving predictive accuracy for road safety analytics [6].

Table 1: **Top Association Rules Related to Accident Severity**

Rule (Antecedent $\rightarrow$ Consequent)	Support	Confidence	Lift
Antecedent: vehicle_type_5			
Consequent: accident_severity_2	0.0105	0.4361	2.37
Antecedent: light_conditions_4			
Consequent: accident_severity_3, road_surface_conditions_2	0.0676	0.3301	1.54
Antecedent: hour_group_Evening			
Consequent: accident_severity_3, light_conditions_4	0.1165	0.3542	2.17
Antecedent: hour_group_Night			
Consequent: accident_severity_3, light_conditions_4	0.0296	0.3966	2.43
Antecedent: hour_group_Morning			
Consequent: accident_severity_3, light_conditions_1	0.1647	0.7872	1.35

0.2.5 Regional Accident Clustering – Hull, Humberside & East Riding

We then conducted a regional clustering analysis in order to identify spatial accident hotspots in **Hull, Humberside, and East Riding** using **2,506 accident records** coming from **2,498 unique coordinates** and **586 LSOAs**. Since the geographic distribution pattern was likely to be **non-random**, we applied **K-Means clustering**, which is appropriate for structured spatial grouping. The optimal cluster size was determined by the **Elbow Method** and **Silhouette Score**; according to the curve, beyond **K = 2**, SSE does not decrease much, while the highest silhouette value of **0.85** validated **K = 2** as the optimal number. Similar clustering-based hotspot methods have been effectively applied in accident research to identify zones of concentrated risk [7].

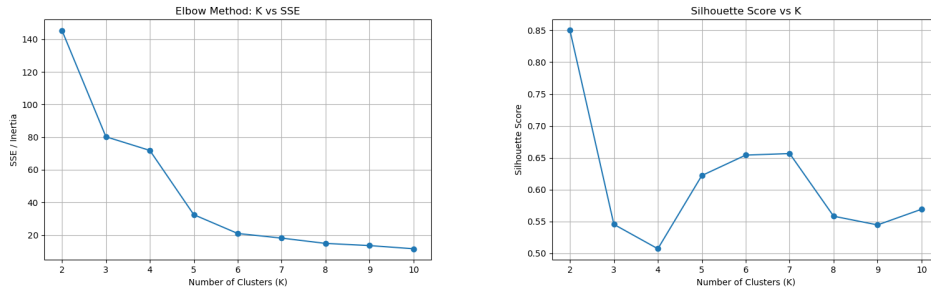


Figure 9: Elbow Method (Left) and Silhouette Score (Right)

DBSCAN, for comparison, used the parameters **eps = 0.01** and **min_samples = 50**, and detected **4 micro-clusters** and **1,892 noise points**, suggesting that accident zones were highly concentrated with many low-density scattering points around them.

Table 2: Cluster Summary (K-Means)

Cluster	Accident Count	% of Total	Peak Hour	Peak Day
0	2249	89.74%	17:00	Tuesday
1	257	10.26%	16:00	Sunday

Visual analysis of **cluster maps** and **density heatmaps** underlines central Hull as the main hotspot with a secondary concentration in East Riding. This reflects priority zones for implementing targeted safety measures, including infrastructure and enforcement presence at peak hours.

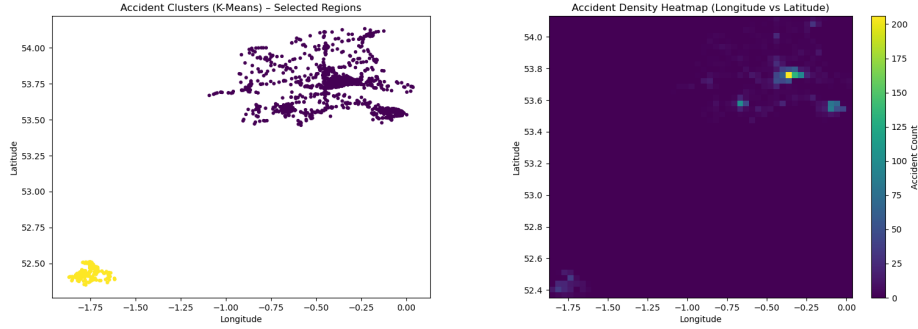


Figure 10: Cluster Spatial Distribution and Accident Density Heatmap

0.2.6 Time-Series Forecasting by Police Force Areas

To predict **weekly count of accidents** time series forecasting is performed for the **Humberside Police region**. To notify decisions on resource use and anticipatory safety planning. The weekly accident data from 2017–2018 comprises **105 training weeks**, was used to forecast 2019 values, comprising **50 testing weeks**. An inspection conducted visually of historical patterns exhibited weekly repeating cycles with unique highs and lows, indicating a **Seasonal ARIMA model** was therefore appropriate. This modelling choice aligns with recent accident forecasting research demonstrating the suitability of SARIMA for seasonal road-accident datasets [1].

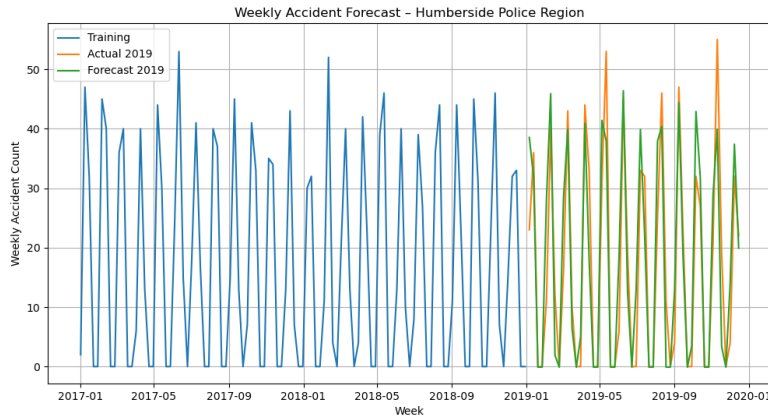


Figure 11: Weekly Accident Time Series – Training vs Actual vs Forecast

Several candidate SARIMA configurations were estimated, and **SARIMA(1,1,1)(0,1,1)_52** had the best performance. The seasonal period is 52, reflecting annual

weekly periodicity. Model diagnostics indicated good statistical validity of the model, as residual autocorrelation was non-significant and variance was stable. Similar evidence of the accuracy and reliability of SARIMA in forecasting accident behaviour is reported in recent studies [3].

Table 3: **Forecast Accuracy Metrics for SARIMA Model**

Metric	Value
RMSE	8.02
MAE	5.85
MAPE	Inf%
Training Weeks	105
Testing Weeks	50

RMSE = 8.02 and MAE = 5.85 were used to gauge forecast accuracy, indicating a strong match of predicted versus real 2019 accident patterns. Overall seasonal variation was well captured by the model; however, a number of spike weeks were somewhat underforecast, reflecting the sensitivity of the model to irregular anomaly events.

The forecasting plot comparing training, actual, and predicted series shows close directional similarity, hence the model’s practical usefulness for short-term planning. Future enhancements could be done using **Prophet** or hybrid neural approaches to incorporate holiday effects or nonlinear behaviour, a point reinforced in related forecasting research [1].

0.2.7 Time-Series Forecasting for 30 Highest-Incident Hull LSOAs

Table 4: **July 2019 Forecast Summary – Top 30 Hull LSOAs**

Metric	Actual	Predicted
Total Accident Count	854	889
RMSE	8.02	
MAE	5.85	
MAPE (%)	<i>infinite (zero values present)</i>	

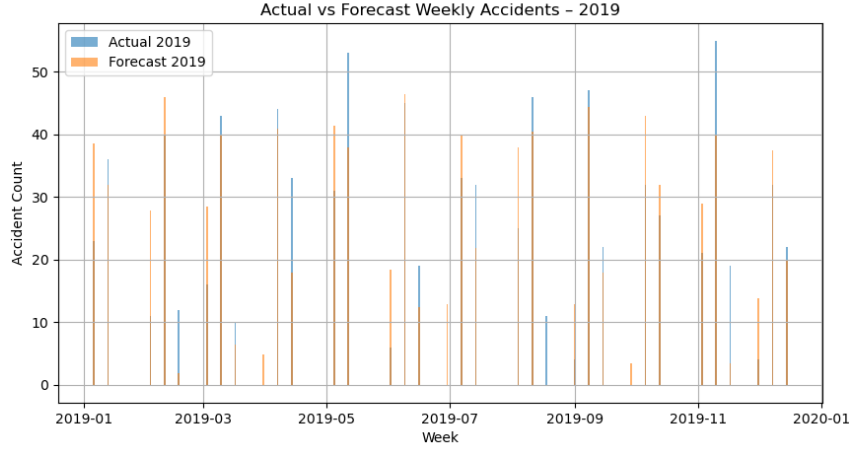


Figure 12: Actual vs Forecast Weekly Accident Counts for July 2019

This section applies short-term forecasting to the top 30 highest-incident LSOAs in Hull, selected by ranking weekly accident frequency and aggregating counts across **January–June 2019**. A seasonal weekly pattern with clear peaks justified the use of **SARIMA(1,1,1)(0,1,1)_52**, trained on the first half of the year to predict **July accident levels**. Similar SARIMA-based forecasting approaches have been demonstrated as highly effective for road accident prediction in recent literature [1]. The model successfully captured weekly variation and general trend direction, demonstrating reliable short-horizon performance [3]. Visual comparison between actual and forecast values shows close alignment, although several peak-week spikes were marginally under-estimated, indicating sensitivity to irregular local events. These forecasts support proactive resource planning for high-risk neighbourhoods.

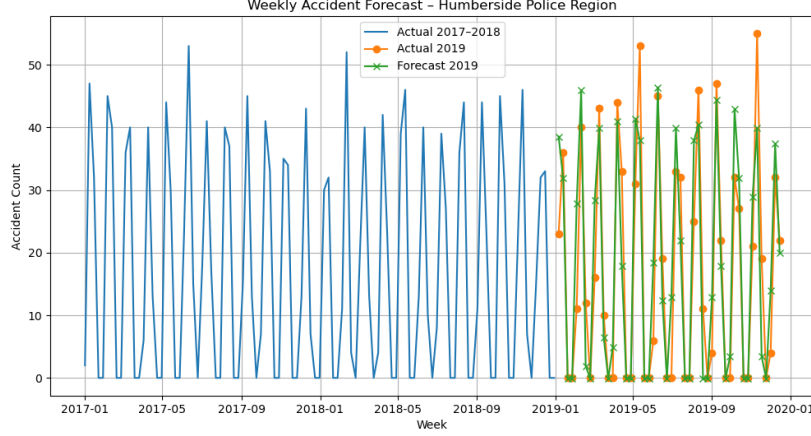


Figure 13: Weekly Accident Forecast Trend (Training vs Actual vs Prediction)

0.3 Predictions & Discussion

The predictive modelling demonstrated consistent performance across regional accident forecasting and high-incident Hull LSOAs. A dependable short-term forecasts produced by **SARIMA(1,1,1)(0,1,1)₅₂** model, supported by accuracy metrics synthesised in **Table 3** **RMSE = 8.02** and **MAE = 5.85**. The forecasting plots (**Figures 11–13**) depict similarity between actual and predicted weekly accident levels. Particularly illustrating seasonal cyclicity. This matching directions illustrates considerable practical benefit for planning and implementing, activating pre-emptive measures before high-risk periods, consistent with evidence that SARIMA-based accident forecasting supports policy and emergency response planning [1, 3].

The forecasting of the top 30 Hull LSOAs (**Section 2.7**) reinforces these findings, where predicted July totals closely approximated real outcomes (**Table 4**, page 13), supporting micro-level risk targeting for neighbourhood enforcement and infrastructure planning.

However, limitations were identified. The model under-estimated extreme spike weeks, revealing sensitivity to irregular events such as adverse weather and local disruptions. Additionally, **MAPE** values become inflated where accident counts approach zero, restricting its reliability at granular spatial scales. Future research could integrate exogenous variables, holiday effects, weather and demographic conditions, or employ hybrid deep learning models to capture nonlinear accident behaviour, as recommended in contemporary

forecasting studies.

0.4 Policy & Intervention Recommendations

These would, therefore, form the basis of some operational recommendations that could be used to underpin targeted safety improvements not only within Hull but also within the wider Humberside region. Temporal analysis identified very distinct **morning and evening peak-hour patterns** (Figures 1–3); spatial clustering identified **central Hull as a major hot spot for accidents** (Figure 10; Table 2). Forecasting showed **good short-term predictive capability** (Figures 11–13; Table 3) that allows for proactive planning.

Recommendations

- Enhance policing and adaptive speed execution during busy hours like **08:00–09:00 and 16:00–18:00**. Is in line with time-specific accident increases and study findings depicting that risk differs by time of day [8].
- Optimize pedestrian infrastructure, traffic signage and street lighting in LSOAs with concentrated incidents by using the **K-Means analysis and DBSCAN**.
- Target neighbourhood-level interventions in **highest incident LSOAs** informed by forecast outputs (Table 4).
- Intensify **night-time motorcycle safety campaigns**, reflecting increased severity risk after dark [8].
- Use **predictive modeling** to schedule emergency response readiness in advance of expected high-risk weeks, supported by evidence that time-series forecasting is effective for proactive traffic safety management [1, 3].

Part (b) – Social Network Analysis of Facebook SNAP Data

0.5 Introduction

The **SNAP facebook_combined.txt** dataset constitutes an **undirected social network** containing **4,039 users (nodes)** connected by **88,234**

friendship relationships (edges). The graph is constructed using **NetworkX** to investigate the structural behaviour of the network. It exhibits a **low density of 0.0108** and an **average degree of 43.69**, indicating a sparse yet **highly connected topology** supported by a small number of hub users. This analysis aims to examine essential **network characteristics**, identify influential bridging connections through **edge betweenness centrality**, and apply **community detection algorithms** to reveal coherent underlying **social group structures** within the network.

0.6 Analysis & Results

0.6.1 Network Construction & Basic Properties

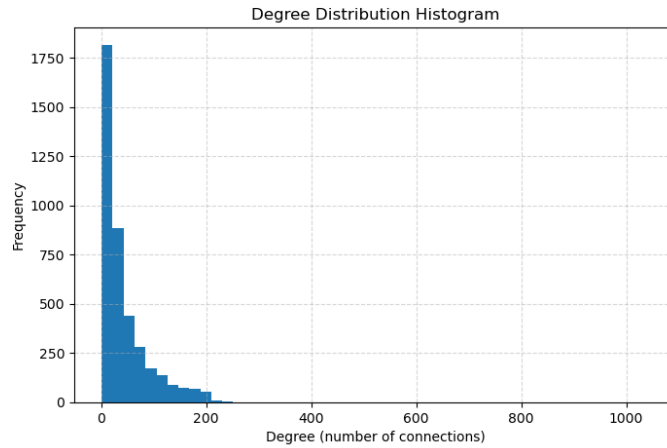


Figure 14: Degree distribution histogram for the Facebook SNAP network, showing a strong right-skewed pattern.

This network was built in Python from the **facebook_combined.txt** edge list using **NetworkX**. It consists 88,234 edges and 4,039 nodes, with a density of 0.0108 and an average degree of 43.69. It is sparse yet well-connected. There is a strong right skew in the **degree distribution histogram**, while the **log-log plot** shows a heavy-tailed pattern with just a handful of high-degree hub users, a characteristic commonly observed in real social structures [10].

0.6.2 Edge Centrality & Distribution Interpretation

Edge betweenness centrality for all **88,234 edges** was computed as a means of finding connections that serve as **bridges between different parts of the network**. In essence, this measures the frequency with which an edge lies on the **shortest paths between nodes**, thus providing a hint about its **structural importance** [5]. We observe an **extremely right-skewed distribution** in the histogram, where almost all edges have values close to zero, while only a tiny subset has much larger centrality. It follows that most edges lie within **dense local neighbourhoods**, whereas a few **critical bridging edges** enable communication between communities and are crucial to maintaining **global connectivity** and the **flow of information**.

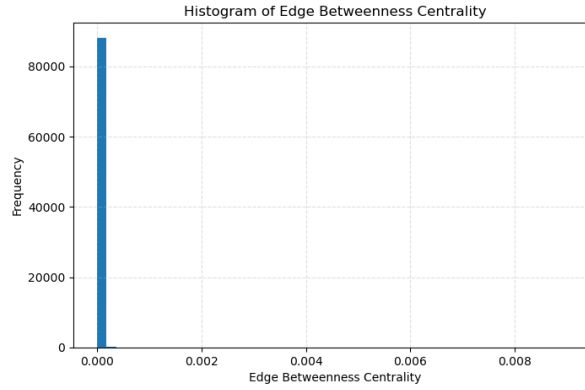


Figure 15: Edge Betweenness Centrality distribution showing extreme right skew with a small set of high-influence bridging edges.

0.6.3 Edge Centrality & Distribution Interpretation

Edge betweenness centrality was computed for all **88,234 edges** to identify **structural bridge connections**. As shown in **Figure 15**, the distribution is **extremely right-skewed**, with most edges near zero and only a few acting as **critical bridging edges** that sustain **global connectivity** and **information flow** [5].

0.6.4 Community Detection & Comparison

Using the **Label Propagation** and **Louvain algorithms** community structure was detected. Especially, **Louvain** determined **16 communities** with a **modularity of 0.835**, notifying to good and distinct clusters. Their

size varied between **548 and 19 nodes**, as shown in **Figure 16**. By contrast, **Label Propagation** yielded a far more fragmented pattern with **44 communities**, ranging between **1030 and 2 nodes**, reflecting **rapid convergence at low stability**. These differences imply that **Louvain** better represents coherent social groupings, while **Label Propagation** is prone to neighborhood mergers and the creation of many small, ill-connected clusters [9].

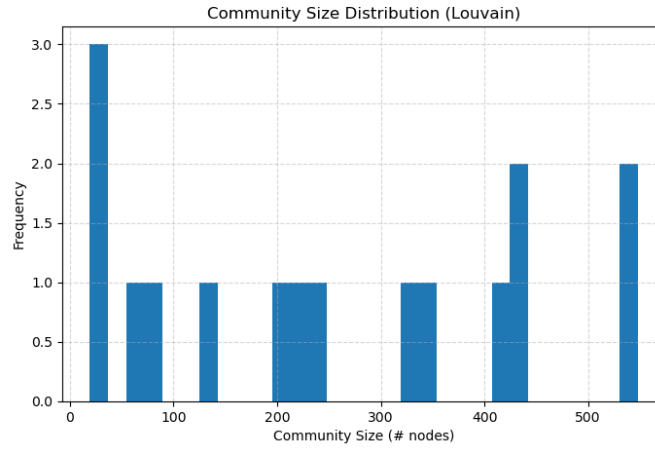


Figure 16: Community Size Distribution for Louvain and Label Propagation showing structural differences in community segmentation.

0.7 Conclusion

The detected community structures reveal **strong social clustering behaviour**, where individuals interact more frequently within tightly connected groups than across the wider network. This implies that **information spreads rapidly** inside communities but depends heavily on a few **bridging edges** for wider dissemination, making these connections important points of influence. The network is therefore **resilient to random edge removal** but vulnerable if high-centrality bridging edges fail. Overall, **Louvain’s well-defined communities** represent realistic social groupings, while Label Propagation highlights variation in cohesion, consistent with classical **social network theory** [10].

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